

DRAFT EMPR: WALMER HOUSING DEVELOPMENT ON ERF 11305

PROJECT NAME:

THE PROPOSED CONSTRUCTION OF WALMER HOUSING DEVELOPMENT ON ERF 11305, PORT ELIZABETH
WITHIN THE NELSON MANDELA BAY MUNICIPALITY, EASTERN CAPE PROVINCE

DEDEAT REFERENCE: ECm1/C/LN2/M/07-2026

REPORT TITLE:

DRAFT ENVIRONMENTAL MANAGEMENT PROGRAMME

DATE: 05 AUGUST 2026

ENVIRONMENTAL ASSESSMENT PRACTITIONER:



CONSULTING ENGINEER:



APPLICANT:



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DOCUMENT REVISION STATUS

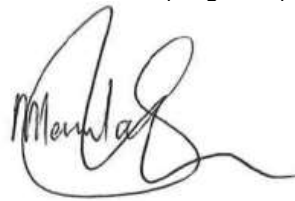
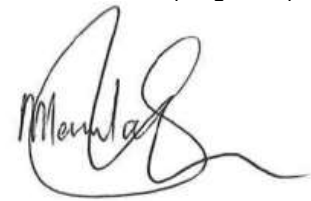
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DATE	July 2026	July 2026	August 2026

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TERMS AND DEFINITIONS

TERMS	DEFINITION
AUDIT	Regular inspection and verification of Construction activities for implementation of the approved EMP
CONSTRUCTION ACTIVITY	A Construction activity is any action taken by the contractor, his subcontractors, suppliers or personnel during the Construction process.
CONTRACTOR	Construction companies are appointed on behalf of the client to undertake the Construction activities, as well as their subcontractors and suppliers.
DWS	Department of Water and Sanitation.
EMERGENCY SITUATION	An incident, which potentially has the ability to significantly impact on the environment, and which, could cause irreparable damage to sensitive environmental features. Typical situations entail amongst others the:- • Spill of petroleum products and lubricants into the aquatic system; • Potential damage, erosion and slumping of unstable river embankments or drainage channels; • Potential event of impeding the continuous flow of water to downstream water users dependent on the flow.
EMPR	Environnemental Management Programme Report. The EMPR for the project sets out general instructions that will be included in a contract document for the Construction phase of the project. The EMPR will ensure the Construction activities are conducted and managed in an environmentally sound and responsible manner.
ENVIRONMENT	The environment means the surroundings within which humans exist and that could be made up of water, air, soil, sand, plants and animals.
EAP	Environmental Assessment Practitioner.
ENVIRONMENTAL IMPACT	An impact or environmental impact is the change to the environment, whether desirable or undesirable, that will result from the effect of a Construction activity. An impact may be the direct or indirect consequence of a Construction activity.
ENVIRONMENTAL CONTROL OFFICER	A qualified person nominated by the appointed contractor and/or client who will ensure the day-to-day implementation of the EMP by contractors during Construction and operation of the project.

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TERMS	DEFINITION
ENVIRONMENTAL MANAGEMENT PROGRAMME (EMPr)	A detailed plan of action prepared to ensure that recommendations for enhancing or ensuring positive environmental impacts and limiting or preventing negative environmental impacts are implemented during the life-cycle of the project. This EMPr focuses on the construction phase, operation (maintenance) phase and decommissioning phase of the proposed project.
GENERAL WASTE	Domestic waste, commercial waste, non-hazardous industrial waste and builders rubble e.g. paper, plastics, food, tins, wood, etc.
HAZARDOUS WASTE	It is any waste that contains organic or inorganic elements or compounds that may, owing to the inherent physical, chemical or toxicological characteristics of that waste, have a detrimental impact on health and the environment.
HERITAGE RESOURCES	This means any place or object of cultural significance, including all human-made phenomena and intangible products that are the result of the human mind. Natural, technological or industrial features may also be part of heritage resources, as places that have made an outstanding contribution to the cultures, traditions and lifestyles of the people or groups of people of South Africa
HERITAGE	That which is inherited and forms part of the National Estate (historical places, objects, fossils as defined by the National Heritage Resources Act, 1999 (Act No. 25 of 1999).
IEM	Integrated Environmental Management.
IMPACT	A description of the potential effect or consequence of an aspect of the development on a specified component of the biophysical, social or economic environment within a defined time and space.
NEMA	National Environmental Management Act.
REHABILITATION	Rehabilitation is defined as the return of a disturbed area, feature or structure to a state that approximates to the state (where possible) that it was prior disruption, or to an improved state.
PALAEONTOLOGY	Any fossilised remains or fossil trace of animals or plants which lived in the geological past, other than fossil fuels or fossiliferous rock intended for industrial use, and any site which contains such fossilised remains or trace.
POLLUTION	The National Environmental Management Act, 1998 (Act No. 107 of 1998) defined pollution to mean any change in the environment caused by – substances; radioactive or other waves; or noise, odours, dust or heat emitted from any activity, including the storage or treatment of waste or substances, construction and the provision of services, whether engaged in by any person or an organ of state, where that change has an adverse effect on human health or well-being or on the composition, resilience and productivity of natural or managed ecosystems, or on materials useful to people, or will have such an effect in the future.

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TERMS	DEFINITION
PROJECT/SITE MANAGER	A person who represents the client and is responsible for enforcing the technical and contractual requirements of the project.
SAHRA	South African Heritage Resource Agency
SOLID WASTE	All solid waste, including Construction debris, chemical waste, excess cement/concrete, wrapping material, timber, tins and cans, drums, wire, nails, food and domestic waste (e.g. plastic packets and wrappers).
TOPSOIL	The layer of soil covering the earth which provides a sustainable environment for the germination of seeds, allows water penetration, and is a source of micro-organisms and plant nutrients.
WASTE	: Waste means any substance, whether or not that substance can be reduced, re-used, recycled and recovered • that is surplus, unwanted, rejected, discarded, abandoned or disposed of; • which the generator has no further use of for the purposes of production; • that must be treated or disposed of; or • that is identified as a waste by the relevant Minister by notice in the Gazette, and includes waste generated by the mining, medical or other sector, but— • a by-product is not considered waste; and • any portion of waste, once re-used, recycled and recovered, ceases to be waste.
WASTE DISPOSAL FACILITY	Waste disposal facility means any site or premise used for the accumulation of waste with the purpose of disposing of that waste at that site or on that premises.
WATER POLLUTION	The National Water Act, 1998 (Act No. 36 of 1998) defined water pollution to be the direct or indirect alteration of the physical, chemical or biological properties of a water resource so as to make it – less fit for any beneficial purpose for which it may reasonably be expected to be used; or harmful or potentially harmful to the welfare, health or safety of human beings; to any aquatic or non-aquatic organisms; to the resource quality; or to property.
WATERCOURSE	A natural channel or depression in which water flows regularly or intermittently in which water flows regularly or intermittently. Or a wetland, lake or dam into which, or from which, water flows; and any collection of water which the Minister may, by notice in the Gazette, declare to be a watercourse as defined in the National Water Act, 1998 (Act No. 36 of 1998).
WUL	Water Use License.

APPLICANT

Details of the applicant are presented in **Table 1** below.

Table 1: Applicant details

Applicant	Nelson Mandela Bay Municipality
Contact Person on behalf of Applicant	Mr Tabiso Mfeya
Physical Address	17 th Floor, Lillian Diedericks Building, Govan Mbeki Ave, 6001
Postal Address	17 th Floor, Lillian Diedericks Building, Govan Mbeki Ave, 6001
Telephone	041 506 2381
Fax	
Email Address	tmfeya@mandelametro.gov.za sstemele@mandelametro.gov.za

ENVIRONMENTAL ASSESSMENT PRACTITIONER

As per the requirements of the NEMA Regulations and in terms of Regulation 14 of Section 24H Registration Authority Regulations, 2016, as amended, the details and expertise levels of the EAP who prepared the report are provided in **Table 2** below.

Table 2: Environmental Assessment Practitioner details

Contact Details	
Consultant	Abantu Environmental Services (Pty) Ltd
EAP	Sive Mlamla
SACNASP Registration	118495 – Environmental Sciences
EAPASA Registration	2022/5204
Cell	078 207 8278
Postal Address	41 Avon Place Overbaakens Gqeberha, 6070
Fax	086 685 9536
Email	info@abantuenvironmental.co.za
Website	Website: www.abantuenvironmental.co.za

Expertise of the Lead EAP are entailed in **Table 3**.

Table 3: Lead EAP Profile

Environmental Assessment Practitioner:	Mr Sive Mlamla
Expertise:	Sive Mlamla is an environmental scientist with 11 years of experience in scientific research, lecturing and environmental consulting. His key expertise includes environmental impact assessments, environmental compliance auditing, waste planning, proposal & business development, project management, strategic planning, research design & execution, remote sensing & GIS applications for land use/cover change mapping and fluvial geomorphology. Qualifications • M.Sc. Geography (Catchment Hydrology) • B.Sc. Honours Environmental Geography • B.Sc. Environmental Sciences (Botany and Geography) Professional registrations: • Registered Professional Natural Scientist (Pr.Sci.Nat) - Reg. No. 118495 – SACNASP • Registered Environmental Assessment Practitioner (Reg. EAP (EAPASA)) - Reg. No. 2022/5204 Training: • Practical Implementation of Environmental Auditing and Monitoring (2025) • WET-Health Tool & WET-EcoServices for Level 1B and Level 2 assessment (2025) • WET-Health Tool for Level 1A assessment (2025) • Environmental Impact Assessments (2025) • Introduction to Synthetic Aperture Radar and Applications (2024) • Earth Observation Data Analysis with Google Earth Engine (2023) • People Management Skills (2017) • Biodiversity spatial data management, acquisition, and analysis (2018) • Water Use Licence Applications via e-WULAAS (2020) • Remote Sensing (land use/cover classification) and SWAT hydrologic modelling (2017) Registrations and memberships: • Member of the South African Wetlands Society Reg. No. S9BPP477 • Member of the Southern African Association of Geomorphologists • Institute of Waste Management of Southern Africa (IWMSA) Reg No. 40120001

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As per the requirements of the NEMA Regulations and Regulation 14(6) in the S24H Registration Authority Regulations, 2016, as amended, details of the Candidate EAP who assisted in the compilation of the report under the supervision of the EAP are provided in **Table 4** below.

Table 4: Candidate Environmental Assessment Practitioner details

Candidate Environmental Assessment Practitioner:	Ms Zimkita Dalasile
Expertise:	Junior Environmental Scientist with 4 years' experience in environmental consulting and scientific research. My expertise includes environmental management, water resource management, water quality management, aquatic ecosystems, environmental impact assessments, scientific report writing, data analysis, environmental monitoring and environmental compliance audits. Qualifications • MSc. Environmental Management with Ecological Water Requirements • Postgraduate Diploma Integrated Water Resource Management • BSc. Environmental and Water Science Professional registration: • Candidate Environmental Assessment Practitioner (Cand. EAP (EAPASA)) - Reg. No. 2025/20366 • Candidate Natural Scientist (Cand.Sci.Nat.) - Reg. No. 132501 – SACNASP Training: • Introduction Environmental Impact Assessments (2024)

DECLARATION OF INDEPENDENCE

I, **Sive Mlamla**, in my capacity as an Environmental Assessment Practitioner, hereby declare that I –

- Act as an independent consultant
- Do not have any financial interest in the undertaking of the activity, other than remuneration for the work performed
- Have and will not have vested interest in the activity
- Have no, and will not engage in, conflicting interests in the undertaking of the activity
- Undertake to disclose any material information that has or may have the potential to influence the decision of the competent authority or the objectivity of any report, plan, or document
- Will provide the competent authority with access to all information at my disposal regarding the report, whether such information is favourable to the Client or not
- Based on information provided to me by the Client and in addition to information obtained during the course of this study, have presented the results and conclusion within the associated document to the best of my professional ability
- Reserve the right to modify aspects pertaining to the present investigation should additional information become available through on-going research and/or further work in this field
- Undertake to have my work peer reviewed on a regular basis by a competent specialist.



Mr Sive Mlamla (Pr. Sci.Nat, Reg. EAP (EAPASA))
Environmental Assessment Practitioner

1. INTRODUCTION

Abantu Environmental Services (Pty) Ltd has been appointed as the independent Environmental Assessment Practitioners in terms of the National Environmental Management Act (Act No.107 of 1998, NEMA) to conduct the EIA process for applying for the relevant authorisations. The proposed project entails the construction of the Walmer housing development (~43 ha), including associated facilities and infrastructure on Erf 11305, Walmer. The erf is located west of the existing Wamer township along Victoria Drive.

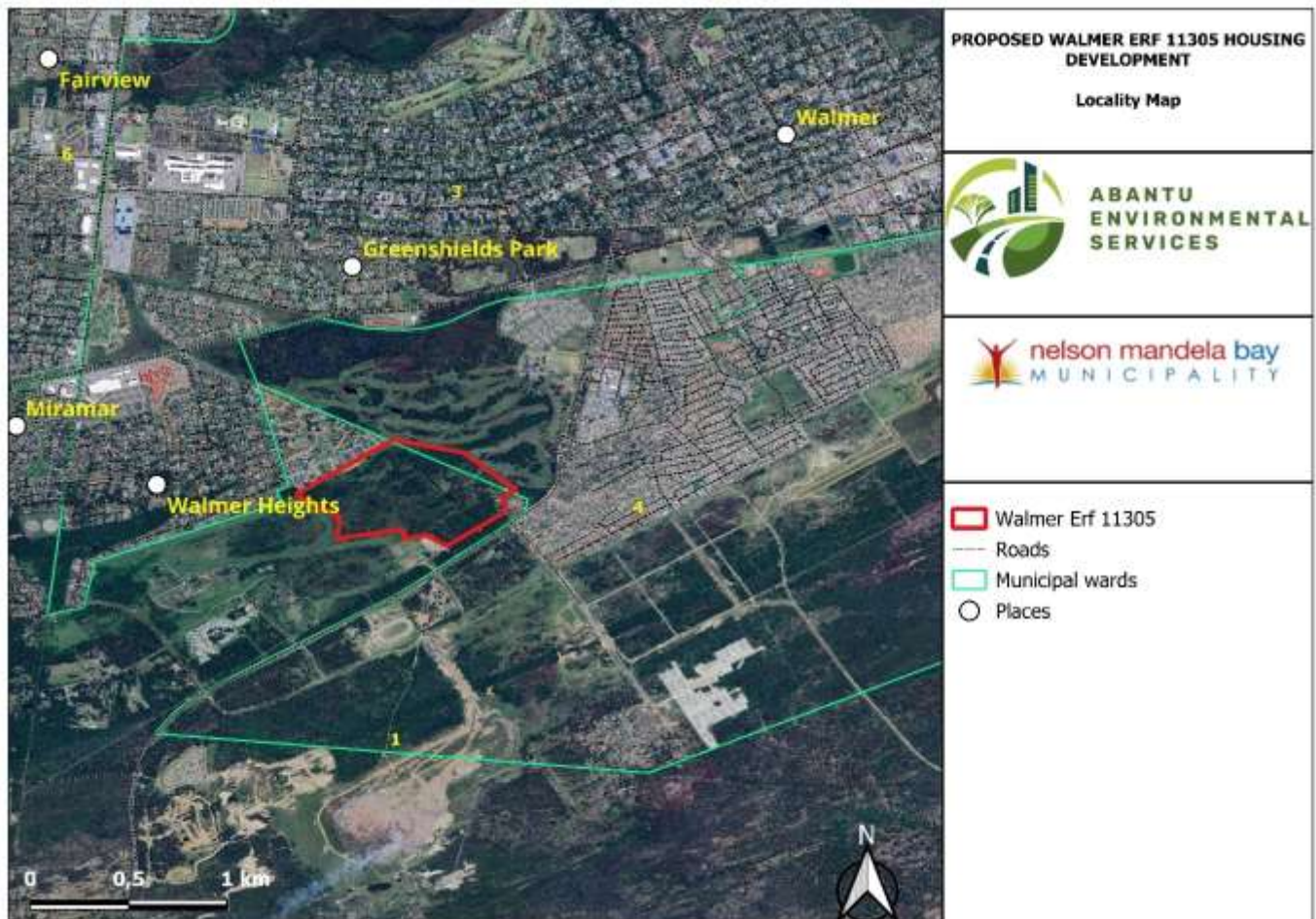


Figure 1. Walmer housing development locality map

The CEMPr has been compiled to ensure that the development is undertaken in compliance with the requirements set out in Section 28 of the National Environmental Management Act (Act No. 107 of 1998) [NEMA] and other relevant legislation under which the Duty of Care Principal is described.

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1.1 LOCATION OF THE ACTIVITY

Property details of the site where the proposed development is located are contained in **Table 5**.

Table 5. Property Information

PROJECT ASPECT	DESCRIPTION	
District	Nelson Mandela Bay Municipality	
Local Municipality	Nelson Mandela Bay Municipality	
Registration Division	Port Elizabeth	
Wards	1	
Development footprint size	43.73 ha	
Coordinates	Latitude (S)	Longitude (E)
	33° 59'42.69"	25° 34'9.21"
	33° 59'45.02"	25° 34'22.98"
	33° 59'52.52"	25° 34'33.44"
	33° 59'53.75"	25° 34'30.98"
	33° 59'56.72"	25° 34'31.46"
	34° 0'3.50"	25° 34'19.84"
	34° 0'1.39"	25° 34'15.70"
	34° 0'2.10"	25° 34'11.54"
	34° 0'0.03"	25° 34'11.03"
	34° 0'2.20"	25° 33'58.14"
	33° 59'58.03"	25° 33'58.67"
	33° 59'53.31"	25° 31'49.3"
Erf number	Erf 11305	
Surveyor General 21 Digit Code	C059003800130500000	
Nearest City	Port Elizabeth	

1.2 DEVELOPMENT DESCRIPTION

The housing development will provide a mix of housing typologies and associated facilities, which include the following:

Housing typologies and associated facilities

- **Free Basic House/ Reconstruction and Development Programme (RDP)**
 - Fully State Subsidised Housing – for beneficiaries earning up to R3 500 per month;
 - Each unit at least 40 m², and costing approximately R160 000 each to build;
 - Beneficiaries will depend entirely on being housed by the state without any expectation of making financial contributions towards the house/services/ transfer/ registration costs for the property to be received; and
 - Units will be either free standing or semi-detached single storey buildings.
- **GAP/ First Linked Individual Subsidy Programme (FLISP)**
 - Partially subsidised housing, where the state subsidy is supplemented by private funding;
 - Each unit >40 m²;
 - For financially employed individuals who can afford mortgage loans of up to R300 000; and
 - Units will be detached, semi-detached single storey or double storey buildings.
- **Social Housing**
 - Units offered for rent to beneficiaries earning between R1 500 and R15 000 per month. Policy stipulates that the rentals paid should not exceed 30% of the gross income of the tenant. This would determine the size of unit allocated to the beneficiary. The units will be owned and managed by an accredited Social Housing Institution that will hold the stock for a minimum of 15 years, and may either re-finance for another 15 years or sell it off to tenants thereafter; subject to providing remaining tenants who still wish to rent with equivalent rental (social) accommodation for another period of 15 years
 - Each unit ±30-54 m² and one to two bedroom apartments; and
 - Units will be in 3 or 4 storey apartment buildings, in an access controlled complex, similar to those in the nearby Walmer Link development.
- **Open/GAP market housing**
 - Stand-alone units priced at above R400 000, for beneficiaries earning above R15 000 per month, with those properties adjacent to Walmer Heights and north of the 12 m internal road providing future connection to Arlington Race Course property being designated for development to a minimum value of R620 000.

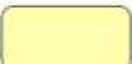
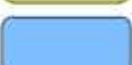
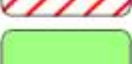
Other land uses (see Table 6)

- Business Zone I- to provide for business's such as a supermarket, bottle store, service trades etc with parking provision at 3.33 bays per 100m².
- Community Zone II- school site
- Authority Zone- authority uses such as police station, fire services, telecommunications facilities etc.
- Open Space Zone I- public open space
- Open Space Zone III - conservation usage
- Transportation Zone I

The total area of the site is 43.73 ha. The Gross Residential Area (single dwellings) is 11.11 ha with Gross Residential Density (single dwellings) being 14.90 units/ha and a Nett Residential Density (single dwellings) being 58.69 units/ha. The average erf size for single dwellings is 170.41m². The estimated number of residential opportunities amounts to 952 units, constituted by single dwellings (652 units) and high density residential (300 units).

Zoning detail (**Table 6**). The design and typologies the current application remain the same to those previously authorised.

Table 6. Proposed development zoning information

Zoning	No. Erven	Area in m ²	% Allo
 General Res Zone 4	1	24236.23 m ²	5.54%
 Single Res Zone 1 (LCH)	509	81629.13 m ²	18.66%
 Single Res Zone 1 (Gap)	143	29478.47 m ²	6.74%
 Business Zone 1	1	11763.94 m ²	2.69%
 Community Zone 2	1	15598.56 m ²	3.57%
 Authority Zone	2	3659.82 m ²	0.84%
 Open Space Zone 1	11	128761.40 m ²	29.44%
 Open Space Zone 3 (Conservation Usage)	13	76015.22 m ²	17.38%
 Transport Zone 1	-	66130.23 m ²	15.15%
Total	681	437273.00 m²	100.00%

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Figure 2. Proposed project layout

1.3 STRUCTURE OF THE EMPR

For the holistic approach to the management of environmental impacts during the construction of the proposed Walmer housing development, this EMPr sets out the methods by which proper environmental controls are to be implemented by the Contractor and all other parties involved. The EMPr is divided into four phases of development. Each phase has specific issues unique to that period of the construction and operation. The impacts are identified and given a brief description. The phases of the development are identified below:

Table 7. Development phases and their descriptions

Phases	Description
Planning and Design	This section will provide guidelines on Planning and Design activities such as complying to relevant legal environmental and social authorisations; site establishment and security; public safety; environmental induction and training and awareness; site access and health and safety.
Construction	This section will provide guidelines on the construction methods and considerations. Environmental actions, procedures and responsibilities as required during the construction phase are specified. These specifications will form part of the contract documentation and therefore the Contractor will be required to comply with these.

Phases	Description
Rehabilitation	This section of the EMPr provides management principles for the rehabilitation phase of the Development. This will include best practice, procedures and responsibilities as required for various associated activities.
Operation	This section will provide guideline on the practice and responsibility as required for various activities during operation.

2. RECEIVING ENVIRONMENT

2.1 CLIMATE

The weather patterns of the Nelson Mandela Bay area change throughout the day as it lies at the confluence of several climatic regimes, the most important of which are temperate and subtropical (Stone, 1988). Gqeberha is dominated by topographical or gradient winds for most of the year (Grobler, 2012). The area experiences westerly winds throughout the year, though in summer the percentage of easterly winds reached more than 40% (Schumann et al., 1999). Maximum and minimum mean temperatures are experienced in February and July, respectively (McCallum 1981). Exceptionally high temperatures (~30°C) can occur during berg wind conditions that develop frequently in autumn and winter. The mean annual rainfall for the Port Elizabeth area is approximately 600 mm (Stone 1988). The strongest winds occur during October and November, with weakest winds during May and June (Schumann et al. 1991).

2.2 TOPOGRAPHY AND HYDROLOGY

The area is characterized by flats to undulating terrain. Elevation ranges from 104 m to 110 m, with an average slope of 1.9% to 2.4%. The study area is situated within quaternary catchment M20A, located within the Mzimvubu-Tsitsikamma Water Management Area. The site occurs within the catchment area of the Baakens River, however there is no connectivity between the Baakens and the proposed site (Scherman Colloty & Associates, 2014). According to the NFEPA database, a NFEPA artificial wetland is located southeast of the boundary of the site, this feature is a reservoir.

Based on the Aquifer Classification and Aquifer Vulnerability Map, the aquifer is classified as a poor aquifer region which is a low to negligible yielding aquifer system of moderate to poor water quality, with least vulnerability meaning is only vulnerable to conservative pollutants in the long term when continuously discharged or leached.

2.3 GEOLOGY AND SOILS

The proposed area is underlain by coastal aeolianites (ancient, wind-blown dune sands) of the Nanaga Formation (Algoa Group) of Pliocene to Early Pleistocene age. These ancient dune sands crop out extensively to the west and east of Port Elizabeth. In the study area they unconformably overlie Palaeozoic sandstones and quartzites of the Peninsula Formation (Almond, 2014). The Nanaga Formation overlies the Alexandria Formation paraconformably, semi-consolidated, medium-grained calcareous sandstone beds are characterized by large-scale cross-bedding, interpreted as aeolian deposits (Booth et al., 1999). The Nanaga Formation is very prominent as fine-grained red sands.

2.3 TERRESTRIAL AND AQUATIC BIODIVERSITY

2.3.1 TERRESTRIAL

According to the National South African Vegetation Map (2018), the study area falls within both the Eastern Fynbos-Renosterveld Bioregion, within the Fynbos Biome and the Albany Thicket Biome. The vegetation types that occur in this area include the Algoa Sandstone Fynbos and Sardinia Forest Thicket. Algoa Sandstone Fynbos is the dominant vegetation type, covering most of the site, while Sardinia Forest Thicket occurs in a smaller section along the southern boundary. The vegetation types are described below:

(a) Algoa Sandstone Fynbos

Algoa Sandstone Fynbos is a grassy shrubland (graminoid fynbos) on coastal flats between Van Stadens River and Summerstrand in the Gqeberha area. It occurs on acidic lithosol soils derived from Ordovician sandstones of the Table Mountain Group (Cape Supergroup) (SRK Consulting, 2024). This vegetation type occurs on flat to slightly undulating plain and grasses become dominant especially in wet habitats.

The Algoa Sandstone Fynbos is classified as **Critically Endangered (CR)** according to The Revised National List of Threatened Terrestrial Ecosystems –2022.

(b) Sardinia Forest Thicket

This vegetation type is dominated by thicket clumps typical of Algoa Dune Thicket. The matrix is forest characterized by Cape ash *Ekebergia capensis* and coral trees *Erythrina caffra* (SRK Consulting, 2008). The Sardinia Forest Thicket is classified as **Vulnerable (VU)**.

The vegetation on site can be described as sparse to moderate shrubs as well as presence of forest clumps. Based on field observations the site is no longer pristine as it has been disturbed by illegal dumping, grazing, access tracks and informal soccer fields. According to the Nelson Mandela Bay Critical Biodiversity Areas (2009), none of the proposed development footprint falls within any CBAs.

(c) Forest

Forest are protected under the National Forest Act, Act 84 of 1998. A permit is required to disturb forest. According to a forest survey conducted by CEN Integrated Environmental Management Unit (2014), the proposed site occurs predominantly in two major clumps on the northern and south-eastern borders, all forest clumps that were observed were a matrix of forest species interspersed with alien vegetation and pioneer species, and all have protected species in terms of the National Forestry Act.

2.3.2 AQUATIC AND STRATEGIC WATER SOURCE AREAS

ESAs are areas that must be maintained in at least fair ecological condition (seminatural/moderately modified state) in order to support the ecological functioning of a CBA or protected area, or to generate or deliver ecosystem services, or to meet remaining biodiversity targets for ecosystem types or species, when it is not possible or not necessary to meet them in natural or near-natural areas (SANBI, 2017).

The project area is located in an aquatic Ecological Support Area (ESA) 1 according to the 2019 Eastern Cape Biodiversity Conservation Plan. As reported by the national web based environmental screening tool, the aquatic biodiversity sensitivity on the site is very high. Although the aquatic biodiversity sensitivity is rated as high, no wetlands or watercourses were observed within the proposed development footprint.

According to Le Maitre *et al.*, (2018), Strategic Water Source Areas (SWSAs) are areas of land that either: (a) supply a disproportionate quantity of mean annual surface water runoff in relation to their size and are considered nationally important; or (b) have high groundwater recharge and where the groundwater forms a nationally important resource; or (c) areas that meet both criteria (a) and (b). The screening tool flags the development area as being located in a Strategic Water Source Area (Tsitsikamma).

2.4 SOCIO-ECONOMIC ATTRIBUTES

The proposed project is located in Gqeberha (formerly known as Port Elizabeth) located in Nelson Mandela Bay Municipality, in the Eastern Cape Province of South Africa and is one of eight Metropolitan Municipalities. In 2001, the Nelson Mandela Bay Metropolitan Municipality was formed as a single administrative area covering inter alia Gqeberha, Kariega (formerly Uitenhage), Despatch and a number of surrounding areas.

The city has a unique advantage in that it possesses two ports, namely Port Elizabeth Harbour and Ngqura. This presents several opportunities related to the diversification of port activities, the maritime sector, and the distinction between 'dirty' and 'clean' port uses. The NMBM Municipality has a size of 195 890.76 ha and covers the smallest area of land (1952 km²) in the Eastern Cape, which is the second largest province in South Africa (Almond *et al.*, 2008).

2.4.1 POPULATION

The current population of Nelson Mandela Bay is estimated at 1 190 496 (Community Survey, 2022), with a growth rate of 1.54% (IHS, 2017), which is lower than that of other metropolitan areas in South Africa, such as Ekurhuleni (2.1%) and Tshwane (2.6%) people. According to the census the Nelson Mandela Bay Municipality population is split 52.7% to 47.3% in favour of females, with the Black African population accounting for 62.7% of the overall population, Coloureds accounting for 19.2%, Whites are 15.7% and the rest are a mix of different ethnicities. According to the census, the elderly constitute 8.8% of the population while the young children under the 14 year make up 22.3% of the population. The literacy levels in the local municipality by 2016 were measured at 76% of the population, which is a significant increase from the 66% measured in 2006.

2.4.2 EDUCATION

According to the Municipality profile of the Nelson Mandela Bay Metro (2016) there is a total number of 324 schools, which consist of 269 ordinary public schools, 11 special needs schools, 25 independent schools and 19 Early Childhood Development (ECD) Centres within the municipality. A comparison with other metros shows that 75.8% of learners aged 7 to 24 years in Nelson Mandela Bay attended an educational institution as compared to 80.2% in Buffalo City and 79.2% in Mangaung (79.2%). The lowest attendance was observed in Cape Town (69.8%) and eThekweni (71.1%). People 20 years and older with less than Grade 7 as highest level of education are 4.1%. Buffalo City is 11.4% and Cape Town is 6.1%. 0.8% of people 20 years and older in Nelson Mandela Bay have no schooling compared to a Metro average of 1.4% (STATS SA GHS, 2019).

2.4.3 ECONOMY

Whilst the NMBM is undoubtedly an important node of activity within the economy of the Eastern Cape, it is characterized by several challenges in terms of economic development. These include:

- A high unemployment rate (36.6%), compounded by the low education levels of the labour force, including large numbers of illiterate adults with limited employment prospects.
- Ageing and inadequate investment in the maintenance and upgrading of infrastructure.
- The dependence on the automotive sector and insufficient diversification within the manufacturing and others sectors.
- A lack of up-to-date local economic statistics and monitoring and evaluation systems.

STATS SA states that: “according to ECSECC (Eastern Cape Socio Economic Consultative Council), the GDP growth rate for the Nelson Mandela Bay Municipality was 2.1% in 2010 and the GDP per capita R52 147. The largest economic sectors in the Nelson Mandela Metro are manufacturing, finance, community services and transport. Community services, trade and manufacturing sectors are the sectors that create the most employment in the Metro”. Good quality infrastructure is key to sustainable social, economic and industrial development. Poor infrastructure hampers development, growth and ability to trade in the domestic and global economy.

Economic infrastructure, which includes transport, energy, telecommunications, water and sanitation provides services which are of fundamental importance for development. In NMBM it is deficient and this is exacerbated by inadequate maintenance and thus prematurely deteriorating installations and services. Infrastructural services are often overlooked as a means to alleviate poverty and improve environmental conditions. In order for Nelson Mandela Bay to grow and develop a sustainable economy, it is important to build new economic drivers to replace or augment the ones that have served the region in the past.

2.4.4 HOUSING

According to STATS SA General Households Survey (2019), 21,6% of households in Nelson Mandela Bay listed grants as their main source of income. A total of 640,000 individuals, accounting for approximately 49.6% of the overall population, live in poverty within the Nelson Mandela Bay Metropolitan area (ECSECC, 2017). In the 2017/18 financial year, all qualifying households with an income of less than R3 200 per month (equivalent to two state pensions) received access to free basic services provided by the Municipality through its Assistance To The Poor Programme (ATTP). There is a slow pace in housing delivery within the municipality with 92% of households living in formal dwellings, 7.4% live in informal dwellings and the remaining 0.6% live in

traditional and other dwellings. The current housing demand in the metro is estimated at 126 550 (Nelson Mandela Bay Municipality IDP, 2022/23-2026/27).

2.4.5 BASIC SERVICES

(a) Water

According to the NMBM IDP (2022/23-2026/27), the NMBM has extent of wate provision backlog in recently established informal settlements. The municipality has about 99% of households that have access to piped water as follows:

- 82.3% have access to piped water inside dwelling.
- 11.4% have access to piped (tap) water inside the yard.
- 5.4% have access to piped (tap) water on community stand.
- 0.8% have no access to basic water.

(b) Sanitation

About 96.2% of households in the Nelson Mandela Bay Municipality has access to basic sanitation. Basic sanitation includes flush toilets connected to a public sewerage system, a septic tank or a pit toilet with a ventilation pipe. There are approximately 6010 buckets in circulation and 2 834 chemical toilets in informal settlements. The Municipality rolls out basic sanitation in informal settlements and relocates households to formal sites with waterborne sanitation (NMBM IDP 2022/23-2026/27).

(c) Electricity

According to Census 2022, 96.5% of households have access to electricity from the mains electricity supply, 3.4% use other sources of energy for lighting whilst the remaining 0.1% have no access to any source of energy.

3 PURPOSE OF THE ENVIRONMENTAL MANAGEMENT PROGRAMME

The EMPr forms part of the contractual obligations to which all contractors/employees involved in construction, maintenance, or decommissioning work must be committed. It serves as a guideline and baseline information document for the construction and operations of the proposed project and aims to comply with section 24N of the National Environmental Management Act, 1998 (Act No. 107 of 1998), as well as the Environmental Impact Assessment Regulations, 2014 as amended.

This section outlines the environmental management guideline measures as encompassed in **Appendix 4** of the EIA Regulations as amended that will be undertaken to avoid or minimize impacts on the environment throughout all phases of the proposed development.

The content of the EMPr prepared in terms of the Regulations is outlined in **Table 8** below. The relevant sections of the EMPr in which the requirements are addressed, is also outlined.

Table 8. Alignment to Amended NEMA 2014 EIA Regulations, Appendix 4 of GNR 326 of 2017

Prescribed content of an EMPr	Reference in the EMPr
1. (1) An EMPr must comply with section 24N of the Act and include-	
(a) details of –	Before Section 1
(i) the EAP who prepared the EMPr.	
(ii) the expertise of that EAP to prepare an EMPr, including curriculum vitae.	
(b) a detailed description of the aspects of the activity that are covered by the EMPr as identified by the project description.	Section 1.3
(c) a map at an appropriate scale which superimposes the proposed activity, its associated structures, and infrastructure on the environmental sensitivities of the preferred site, indicating any areas that should be avoided, including buffers.	Section 1, Figure 1.
(d) a description of the impact management outcomes, including management statements, identifying the impacts and risks that need to be avoided, managed and mitigated as identified through the environmental impact assessment process for all phases of the development including-	Section 6, 7, 8 & Section 9
(i) planning and design.	
(ii) pre-construction activities.	
(iii) construction activities.	
(iv) rehabilitation of the environment after construction and where applicable post closure.	
(v) where relevant, operation activities.	
(f) A description of proposed impact management actions, identifying the manner in which the impact management outcomes contemplated in paragraphs (d) will be achieved, and must, where applicable, include actions to –	Section 6, 7, 8 & Section 9
(i) Avoid, modify, remedy, control or stop any action, activity or process which causes pollution or environmental degradation.	
(ii) Comply with any prescribed environmental management standards and practices.	
(iii) Comply with any applicable provisions of the Act regarding closure, where applicable.	
(iv) Comply with any provisions of the Act regarding financial provisions for rehabilitation, where applicable.	
(g) The method of monitoring the implementation of the impact management actions contemplated in paragraph (f).	Section 6, 7, 8 & Section 9

Prescribed content of an EMPr	Reference in the EMPr
(h) The frequency of monitoring the implementation of the impact management actions contemplated in paragraph (f).	Section 6, 7, 8 & Section 9
(i) An indication of the persons who will be responsible for the implementation of the impact management actions.	Section 10
(j) The time periods within which the impact management actions contemplated in paragraph (f) must be implemented.	Section 6, 7, 8 & Section 9
(k) The mechanism for monitoring compliance with the impact management actions contemplated in paragraph (f).	Section 11
(l) A program for reporting on compliance, taking into account the requirements as prescribed by the Regulations.	Section 11
(m) An environmental awareness plan describing the manner in which – (i) The applicant intends to inform his or her employees of any environmental risk which may result from their work. (ii) Risks must be dealt with in order to avoid pollution or the degradation of the environment.	Section 4.2 & Section 11
(n) Any specific information that may be required by the competent authority.	Not applicable
1. (2) Where a government notice gazetted by the Minister provides for a generic EMPr, such generic EMPr as indicated in such notice will apply.	Not applicable

The main objectives of this EMPr are therefore to:

- Outline environmental management measures related to project activities and provide project contractors with guidelines for carrying out Construction activities in such a way as to minimize environmental impacts;
- Be used as a foundation for the specific environmental management instructions contained in Construction contract documents, where compliance will be a contractual obligation for contractor(s);
- Be used as an educational tool, for orientation and training of project personnel and contractors;
- To outline functions and responsibilities of responsible persons;
- To state standards and guidelines, which are required to be achieved in terms of environmental legislation;
- To outline mitigation measures and environmental specifications which are required to be implemented for all phases of the project in order to minimize the extent of environmental impacts, and to manage environmental impacts associated with the proposed project; and
- To prevent long-term or permanent environmental degradation.

3.1 ENVIRONMENTAL CODE OF CONDUCT

One of the objectives of the EMPr is to ensure that all the workforce, contractors, sub-contractors, and construction staff understand environmental issues and potential impacts on-site activities. This environmental code of conduct provides the basic rules that must be strictly adhered to.

It is the responsibility of the Site Environmental Officer, the Environmental Officer and ECO (as appointed) to ensure that each contractor, sub-contractor, and workforce understand and adhere to the Code of Conduct.

All persons are obliged to keep to the rules of this code of conduct. Ignorance, negligence, recklessness, or a general lack of commitment resulting in environmental degradation or pollution must not be tolerated.

Environmental Rules

- Do not waste electricity, water, or consumables;
- Only use authorised accesses;
- Do not litter;
- Dispose solid waste to the correct waste containers provided;
- Prevent pollution;
- Use the toilet facilities provided;
- Do not dispose contaminated wastewater to the storm water or the environment;
- Immediately report any spillage from containers, plant, or vehicles;
- Do not burn or bury any waste in the sand;
- Do not trespass onto private properties;
- Strictly leave all animals alone. Never tease, catch, or set devices to trap or kill any animal;
- Never damage or remove any trees, shrubs, or branches unless it forms part of working instructions;
- Do not deface, draw, or cut lettering or any other markings on trees, rocks or buildings in the area;
- Know the firefighting procedure and locations of firefighting equipment; and know the environmental incident procedures.

3.2 CONTRACTUAL OBLIGATION

This EMPr must be included in all tender and contract documentation associated with the development. Note that notwithstanding the Companies Act, 2008 (Act No. 71 of 2008), or the Close Corporations Act, 1984 (Act No. 69 of 1984), the directors of a company or members of a close corporation are jointly and severally liable for any negative impact on the environment, whether advertently or inadvertently caused by the company or close corporation which they represent, including damage, degradation or pollution.

4 MANAGEMENT AND MONITORING PROCEDURE

4.1 ORGANIZATIONAL STRUCTURE AND RESPONSIBILITY

This section indicates the party responsible for implementing the environmental measures and action plans laid out in this EMPr. Formal responsibilities are necessary to ensure that key procedures are executed. Specific responsibilities of the Project Proponent, Project Manager, Site Manager/Engineer, Contractor/Operator and Environmental Control Officer are as detailed below.

Project Developer shall:

- Be fully acquainted with the EMPr for the project;
- Ensure that the Project Engineer and the Contractor/Operator are aware of all specifications, legal constraints, standards, and procedures pertaining to the project specifically with regard to the environment;
- Ensure that all stipulations within the EMPr are communicated and adhered to by the Project Engineer and the Contractor/Operator;
- Monitor the implementation of the EMPr throughout the project by means of regular site visits and meetings; and
- Order the removal of any person(s) and/or equipment in contravention of the specifications of the EMPr.

The Project Manager (Engineer) shall:

- Be fully conversant with the EMPr;
- Ensure compliance with the EMPr;
- Have overall responsibility for the implementation of the EMPr;
- Liaise with the Project Manager and Contractor/Operator on matters concerning the environment;
- Prevent actions that will harm or may cause harm to the environment, and take steps to prevent pollution of the site;
- Implement remedial measures in the event of pollution incidents or environmental impacts;
- Monitor and verify that environmental impacts are kept to a minimum;
- Review and approve construction methods where necessary; and
- Order the removal of any person(s) and/or equipment in contravention of the specifications of the EMPr.

The Contractor shall:

- Be fully conversant with the EMPr;
- Ensure compliance with the EMPr;
- Ensure that all the environmental specifications contained within this EMPr are adhered to at the site;
- Regularly liaise with the Site Manager on matters relating to the environment; and
- Confine activities to the demarcated construction site.

The above responsibilities listed for the Contractor will also apply to any appointed sub-consultants.

The Environmental Control Officer (ECO) shall:

- Be fully conversant with the EMPr;
- Be fully conversant with all environmental legislation and ensure compliance;
- Ensure that all the environmental specifications contained within this EMPr are adhered to at the site;
- Regularly liaise with the Site Manager on matters relating to the environment; and
- Compile monthly reports as to the progress of the construction phases and report to all parties involved (Site Manager, Project Proponent).

4.2 ENVIRONMENTAL AWARENESS TRAINING

Training and environmental awareness is an integral part of a complete EMPr. The overall aim of the training will be to ensure that all site staff is informed of their relevant requirements and obligations pertaining to the relevant authorizations, licences, permits and the approved EMPr and protection of the environment. Environmental awareness training courses should be run for all personnel on site. Two types of courses should be run, one for the Contractor's and Subcontractor's management and one for all site staff and labourers. Courses shall be run in the morning during normal working hours at a suitable venue provided by the Contractor. All attendees shall remain for the duration of the course and sign an attendance register on completion that clearly indicates participant's names, a copy of which shall be handed to the ECO.

- The size of each session shall be limited to 30 people. The Contractor shall allow for sufficient sessions to train all personnel. Subsequent sessions shall be run for any new personnel coming onto site. A Method Statement with respect to the organization of these courses shall be submitted.
- Notwithstanding the specific provisions of this clause, it is incumbent upon the Contractor to convey the sentiments of the EMPr to all personnel and Subcontractors involved with the Works.

3.2.1 Training Course for Management and Foremen

- The environmental awareness training course for management shall include all management staff and foremen. The course, which will be presented by the ECO, will be of approximately one-hour duration.
- The initial course shall be undertaken not less than 7 days prior to commencement of work on site. Subsequent courses shall be held as and when required.

3.2.2 Training Course for Site Staff and Labour

- The environmental awareness training course for site staff and labour shall be presented by the Contractor's SHE Officer from material provided by the ECO unless otherwise required by the Project Specification. The course will be approximately one-hour long.
- The course shall be run not more than 7 days after commencement of work on site with sufficient sessions to accommodate all available personnel. Subsequent courses shall be held as and when required.

3.2.3 Construction Personnel Information Posters

- The Contractor shall erect and maintain information posters for the information of his employees depicting actions to be taken to ensure compliance with the EMPr. Construction personnel information posters shall be laminated and erected in all eating areas, workshops, and site offices. The Contractor shall ensure that the construction personnel information posters are not damaged in any way and shall replace them if any part becomes illegible.
- Examples of these posters will be supplied to the Contractor by the ECO in electronic format.

3.2.4 The Environmental Awareness Training Programme

The environmental awareness training programme should include:

- The induction of all construction and operation staff;
- Signing by all persons, an acknowledgement of receiving and understanding the induction;
- Identification of environmental risks and job specific training on addressing these risks; and training on the implementation of emergency procedures (where necessary).

3.2.5 Topics Covered by the Environmental Awareness Programme

Various topics to be entailed by environmental awareness programmes include by not limited to:

- What is meant by "Environment"?
- Why does the environment need to be protected and conserved?
- How can construction activities impact on the environment?
- What can be done to mitigate against such impacts?
- Awareness of emergency and spills response provisions.

5 LEGISLATIONS AND GUIDELINES

All the applicable environmental standards contained within the environmental legislation will be adhered to. Below are applicable legislations and guidelines for the proposed development and have been identified as applicable:

Legislation/ guidelines	Summary	Legal requirement for this project
NATIONAL		
The Constitution of South Africa, 1996 (Act No.108 of 1996)	The Constitution is the highest and the supreme law in South Africa. The Bill of Rights in chapter 2 section 24 of the Constitution of South Africa Act (Act 108 of 1996) makes provisions for environmental issues and declares that: "Everyone has the right – a) to an environment that is not harmful to their health or well-being; and b) to have the environment protected, for the benefit of present and future c) generations, through reasonable legislative and other measures that: i. prevent pollution and ecological degradation; ii. promote conservation; and secure ecologically sustainable development and use of natural resources while promoting justifiable economic and social development".	The applicant has an obligation to ensure that the project is undertaken in a manner that respects and protects the constitutional rights of all interested and affected parties. The applicant must ensure that the project environment is not harmful and that measures are implemented to prevent pollution so that future generations can enjoy the social and ecological benefits.
National Environmental Management Act, 1998 (Act No.107 of 1998) as amended	The National Environmental Management Act, 1998 (Act No.107 of 1998) (NEMA) is a 'principles based Act' and is an overarching statute regulating various aspects of natural resources use, integrated environmental management and pollution control. The Act provides for the right to an environment that is not harmful to the health and wellbeing of the South African people; sustainable development, environmental protection, equitable distribution of natural resources; and the formulation of environmental management frameworks. Its definition of the environment includes the land and water of the earth, microorganisms, plant and animal life or a combination	The applicant must ensure that construction and operation of activities must be conducted according to the generally accepted principles of sustainable development, integrating social, economic, and environmental factors.

Legislation/ guidelines	Summary	Legal requirement for this project
	of those things, and the inter relationships among them. The Act aims to provide for cooperative environmental governance by establishing principles for decision making on matters affecting the environment, institutions that will promote co-operative governance, and procedures for coordinating environmental functions exercised by organs of state. Section 24 Provides for the prohibition, restriction and control of activities which are likely to have a detrimental effect on the environment.	An application for Environmental Authorisation is being submitted on behalf of the client in line with the requirements of NEMA since the proposed project will trigger listed activities which require authorization prior to commencement. As part of the EIA process, mitigation measures will be proposed to ensure that the significance of the predicted impacts is reduced thus protecting the environment from degradation.
Environmental Impact Assessment Regulation, 2014 as Amended	The Environmental Impact Assessment (EIA) Regulations promulgated under NEMA in 2014 provide a list of activities which are subject to an Environmental Authorisation (EA) process prior to construction or implementation. In accordance with the 2014 EIA Regulations, (as amended) an EIA process is required owing to the applicability of the activities listed in Error! Reference source not found.. According to the NEMA Regulations these activities may not commence without environmental authorization from the competent authority which requires the investigation, assessment and statement of potential impact of activities and must follow the procedure as described in the EIA Regulations. A Basic Assessment (BA) process must be applied to an application if the authorisation applied for is in respect of an activity or activities listed in Listing Notices 1 and/or 3 and a Scoping and Environmental Impact Assessment (S&EIA) process must be applied to an application if the authorisation applied for is in respect of an activity or activities listed in Listing Notice 2. As the Proposed Project includes activities listed in Listing Notice 2 it is necessary that a S&EIA process is undertaken in order for the Eastern Cape Department of Economic Development, Environmental Affairs and Tourism (DEDEAT) to consider the application in terms of NEMA.	An application for Environmental Authorisation is being submitted on behalf of the client in line with the requirements of NEMA EIA Regulations since the proposed project will trigger listed activities which require authorization prior to commencement
National Environmental Management Biodiversity Act (Act No. 10 of 2004)	The National Environmental Management: Biodiversity Act (NEM:BA) makes provisions for achieving the objectives of the United Nations Convention on Biological Diversity, to which South Africa is a signatory. The Bill promotes	The proposed housing development area does not fall within an area identified as a Critical Biodiversity Area (CBA) according to the Eastern Cape Biodiversity Conservation Plan (ECBCP) as well as the NMBMM Bioregional Plan. The proposed

DRAFT EMPR: WALMER HOUSING DEVELOPMENT ON ERF 11305

Legislation/ guidelines	Summary	Legal requirement for this project
	management, conservation and sustainable use of indigenous biological resources, and provides for: • the management and conservation of biological diversity; • the use of indigenous biological resources in a sustainable manner; and • the fair and equitable sharing of benefits arising from the commercialization through bio-prospecting of traditional uses and knowledge of generic resources. The Bill gives effect to international agreements relating to biodiversity which are binding on the Republic and provides for co-operative governance in biodiversity management and conservation and provides for a National Biodiversity Institute to assist in achieving the above objectives. The Act gives wide powers to the National Biodiversity Institute to inter alia protect flora and fauna in appropriate enclosures, the collection of information, undertaking and promotion of research on indigenous biodiversity and the sustainable use of indigenous biological resources, the prevention, control or eradication of listed invasive species, biodiversity planning and other functions. This act lists all critically endangered, vulnerable and protected species. The potential occurrence of any such species will be investigated in the BA process.	development will be within threatened ecosystems as defined by the NEM:BA, namely the critically endangered Algoa Sandstone Fynbos and the vulnerable Sardinia Forest Thicket.
National Water Act, 1998 (Act No.36 of 1998)	In terms of chapter 3 section 12-20, water resources are to be protected, used, developed, conserved, managed and controlled. This Act recognizes that water is a scarce resource; it is a natural resource that belongs to all of South Africa's people. The National Department of Water and Sanitation is responsible for the nation's water resource and also the Minister of Department of Water and Sanitation ensures that the water resource is "protected, used, developed, conserved, managed and controlled" through the implementation of this Act (National Water Act 36 of 1998). This Act makes provisions for the protection of surface water and groundwater resources and their sustainable management for the prevention and remediation of the effects of pollution, and for the control of emergency	The proposed project is not located within 100m of a river or within 500m of a wetland therefore, Section 21 c and i of the National Water Act will not be triggered.

DRAFT EMPr: WALMER HOUSING DEVELOPMENT ON ERF 11305

Legislation/ guidelines	Summary	Legal requirement for this project
	occurrences. Section 21 of the National Water Act (NWA) lists water uses for which a Water Use Licence will be required.	
National Environmental Management: Air Quality Act, 2004 (Act No.39 of 2004)	The objective of the Act is to protect the environment by providing reasonable measures for the protection and enhancement of air quality and to prevent air pollution. The Act makes provision for measures to control dust, and offensive odours. Section 32 of The National Environmental Management: Air Quality Act, 2004 (Act 39 of 2004) deals with dust control measures regarding dust control. The Minister or MEC may prescribe measures for the control of dust in specified places or areas, either in general or by specified machinery or in specified instances, the steps to be taken to prevent nuisance or other measures aimed at controlling dust. The National Dust Control Regulations (2013) provides for the management and monitoring of dust.	The EMPr which will be compiled during the EIA phase must include measures for control of dust during the construction phase, If there are any exceedances observed in terms of the National Dust Regulations then a dust monitoring programme must be submitted to the Competent Authority.
Conservation of Agricultural Resources Act, 1983 (No. 43 of 1983)	This Act provides for the control over the utilisation of the natural agricultural resources of the country in order to promote the conservation of the soil, the water sources and the vegetation and the combating of weeds and invader plants. Section 5 of the Act prohibits the spread of weeds through the prohibition of their sale. GN R1084 (published under CARA) provides categories for the classification of the various weeds and invader plants, and restrictions where these species may occur. Regulation 15E of GN R1084 provides methods to be implemented for the control of weeds and invader species.	CARA finds application throughout the project lifecycle of the Proposed Project. As a result, soil conservation and erosion prevention management and mitigation measures need to be implemented. Furthermore, an Alien Invasives Species Control and Management Plan must be developed and implemented for the duration of the project life cycle of the Proposed Project
Subdivision of Agricultural Land Act, 1970 (Act No. 70 of 1970), as amended	The Subdivision of Agricultural Land Act, 1970 (Act No. 70 of 1970), as amended provides for the subdivision of all agricultural land within the Republic thereby prohibiting certain activities from being undertaken without consent from relevant authority, the Minister of the Department of Agriculture, Land Reform and Rural Development.	This Act could be relevant to the Proposed Project if any portion of land is zoned for agriculture and will need to be leased for a period exceeding 10 years is regulated by the Act
National Forests Act, 1998 (No. 84 of 1998)	The National Forest Act (NFA) empowers the Minister of DFFE to declare and list a tree, group of trees, woodland, or a species of trees as protected. A list of protected tree species is included in GN R908, published in November 2014. Section 7 of the Act prohibits the cutting and disturbance of NFA-listed trees.	A permit may be required for the removal of NFA-listed tree species in terms of Section 4 of the Act, if any will be impacted. Prior to the submission of the permit application to the competent authority, a survey of the project site is required in order to ascertain the presence and distribution of NFA-listed tree species.

DRAFT EMPR: WALMER HOUSING DEVELOPMENT ON ERF 11305

Legislation/ guidelines	Summary	Legal requirement for this project
		There are forest patches within the development area, these need to be avoided as far as possible.
Occupational Health and Safety Act, 1993 (Act No.85 of 1993)	The Occupational Health and Safety Act makes provisions in regulations Section 8 for the general duties of employers to their employees. The act provides for the health and safety of people at work utilising machinery and the protection of others against health and safety risks associated with activities on site/work. General Administrative Regulations (2003) describe the administration of the various OHS Regulations, including the designation of health and safety committees, the reporting and recording of incidents and occupational diseases. This Act is applicable to all contractors during the planning, construction and operational phases of the project. To provide for the health and safety of persons at work and for the health and safety of persons in connection with the use of plant and machinery; the protection of persons other than persons at work against hazards to health and safety arising out of or in connection with the activities of persons at work.	The applicant must ensure that a safe working environment is provided for its employees during construction and operational phases of the project. This includes obtaining the relevant work permits, providing PPE and ensuring all required facilities are available for a working environment that is conducive. All stalls must have adequate training for their various duties and the applicant must ensure that compliance with the OHSA and Construction Regulations is monitored on a regular basis.
Hazardous Substance Act (No 15 of 1973)	This Act regulates the control of substances that may cause injury, or ill health, or death due to their toxic, corrosive, irritant, strongly sensitizing or inflammable nature of the generation of pressure thereby in certain instances and for the control of certain electronic products. To provide for the rating of such substances or products about the degree of danger; to provide for the prohibition and control of the importation, manufacture, sale, use, operation, modification, disposal or dumping of such substances and products. • Group I and II: Any substance or mixture of a substance that might by reason of its toxic, corrosive, etc., nature or because it generates pressure through decomposition, heat, or other means, cause extreme risk of injury etc., can be declared as Group I or Group II substance • Group IV: any electronic product; and • Group V: any radioactive material. The use, conveyance, or storage of any hazardous substance (such as distillate fuel) is prohibited without an appropriate license being in force. It is necessary to identify and list all the Group I, II, III, and IV hazardous substances that may be on the site and in what operational context they are used, stored, or handled. If applicable, a license is required to be obtained from the	Relevant permits must be obtained for the storage of hazardous substances if any will be stored on site during construction. The contractor must ensure that hazardous substances are stored in a safe manner and MSDS are retained on file for all hazardous substances on site.

Legislation/ guidelines	Summary	Legal requirement for this project
National Environmental Management: Waste Act, 2008 (Act No.59 of 2008)	During construction waste will be produced, in either liquid, solid and/or hazardous state, and this waste will be required to be adequately and appropriately disposed of. There are several Regulations or Acts that are applicable to the proposed development in terms of waste management. To reform the law regulating waste management in order to protect health and the environment by providing reasonable measures for the prevention of pollution and ecological degradation and for securing ecologically sustainable development; to provide for institutional arrangements and planning matters; to provide for national norms and standards for regulating the management of waste by all spheres of government; to provide for specific waste management measures; to provide for the licensing and control of waste management activities; to provide for the remediation of contaminated land; to provide for the national waste information system; to provide for compliance and enforcement; and to provide for matters connected therewith.	No authorization is required in terms of NEMWA, however, the applicant must make sure that waste is managed appropriately on site. This includes separation of waste, routine cleanup of the site and spillages as well as disposal at appropriately licensed waste landfills. Where possible, waste should be recycled to minimize volumes of waste disposed to landfills
National Heritage Resources Act (Act No. 25 of 1999)	The protection of archaeological and paleontological resources is the responsibility of a provincial heritage resources authority and all archaeological objects, paleontological material and meteorites are the property of the State. "Any person who discovers archaeological or paleontological objects or material or a meteorite in the course of development must immediately report the find to the responsible heritage resources authority, or to the nearest local authority offices or museum, which must immediately notify such heritage resources authority". According to Section 34 of NHRA, No person may alter or demolish any structure or part of a structure which is older than 60 years without a permit issued by the relevant provincial heritage resources authority. Section 38 Listed Activities:	The site is located in close proximity to a grade IIIb heritage site. A Heritage specialist has been appointed to assess the presence of other heritage resources within the site as well as the significance thereof. The ECPHRA has been identified as a stakeholder in this project and will be provided an opportunity to comment on the findings of the specialist studies.

DRAFT EMPR: WALMER HOUSING DEVELOPMENT ON ERF 11305

Legislation/ guidelines	Summary	Legal requirement for this project
	(a) the construction of a road, wall, power line, pipeline, canal or other similar form of linear development or barrier exceeding 300m in length; (b) the construction of a bridge or similar structure exceeding 50 m in length; (c) any development or other activity which will change the character of a site— (i) exceeding 5 000 m² in extent; or (ii) involving three or more existing erven or subdivisions thereof; or (iii) involving three or more erven or divisions thereof which have been consolidated within the past five years; or (iv) the costs of which will exceed a sum set in terms of regulations by SAHRA or a provincial heritage resources authority; (d) the re-zoning of a site exceeding 10 000 m² in extent; or (e) any other category of development provided for in regulations by SAHRA or a provincial heritage resources authority;	
The Spatial Planning and Land Use Management Act, 2013 (No. 6 of 2013) (SPLUMA)	The Spatial Planning and Land Use Management Act, 2013 (Act No. 6 of 2013) aims to confirm and regulate the role of municipalities in land use planning and management.	Objectives of the Act relevant to the Proposed Project ensure that the system of spatial planning and land use management promotes social and economic inclusion and to provide for the sustainable and efficient use of land.
Housing Act, 1997 (Act No. 107 of 1997)	The Act recognises the Constitutional right to access to adequate housing and clarifies the State's response to this right by setting out the legal plan for the sustainable development of housing. It defines the national, provincial, and local government functions concerning housing development and the financing of housing programmes. According to part 4 Section 9 of the act,	The proposed housing development is in accordance with this Act. The NMBM shall provide residents of Walmer with adequate housing and services such as electricity, water, sanitation etc.

Legislation/ guidelines	Summary	Legal requirement for this project
	every municipality must as part of the municipality's process of integrated development planning take all reasonable and necessary steps within the framework of national and provincial housing legislation and policy to (a) ensure that- (i) the inhabitants of its area of jurisdiction have access to adequate housing on a progressive basis; (ii) conditions not conducive to the health and safety of the inhabitants of its area of jurisdiction are prevented or removed; (iii) services in respect of water, sanitation, electricity, roads, stormwater drainage and transport are provided in a manner which is economically efficient; (b) set housing delivery goals in respect of its area of jurisdiction; (c) identify and designate land for housing development; (d) create and maintain a public environment conducive to housing development which is financially and socially viable; (e) promote the resolution of conflicts arising in the housing development process; (f) initiate plan, co-ordinate, facilitate, promote and enable appropriate housing development in its area of jurisdiction; (g) provide bulk engineering services, and revenue generating services in so far as such services are not provided by specialist utility suppliers; and (h) plan and manage land use and development	

Legislation/ guidelines	Summary	Legal requirement for this project
Integrated Environmental Management Information Guidelines Series:	This series of guidelines was published by the Department of Environmental Affairs (DEA) and refers to various environmental aspects. Applicable guidelines in the series for the proposed project include: ● Guideline 5: Companion to NEMA EIA Regulations, 2010; ● Guideline 7: Public participation; and ● Guideline 9: Need and desirability. Additional guidelines published in terms of the NEMA EIA Regulations, 2014 (as amended), in particular: ● Guideline 3: General Guide to EIA Regulations, 2006; ● Guideline 4: Public Participation in support of the EIA Regulations, 2006; and Guideline 5: Assessment of alternatives and impacts in support of the EIA Regulations, 2006.	These guidelines have been consulted in the compilation of this report as well as the public participation process that will be undertaken.
Municipal Systems Act (Act 32 of 2000)	The Municipal Systems Act provides for the core principles, mechanisms and processes that are necessary to enable municipalities to provide for community participation and for the integration of all activities for the overall social and economic upliftment of communities in harmony with their local natural environment. It also states that a fundamental aspect of the new local government system is the active engagement of communities in the affairs of municipalities of which they are an integral part. The Act requires the implementation and monitoring of Integrated Development Plans, the setting of targets and key performance indicators, including environmental targets, as well as the preparation of by-laws and policies that deal with environmental issues.	The NMBM has been included as an I&AP for this project and the municipal IDP has been consulted in compilation of this report.
PROVINCIAL LEGISLATION AND POLICY FRAMEWORK		

DRAFT EMPR: WALMER HOUSING DEVELOPMENT ON ERF 11305

Legislation/ guidelines	Summary	Legal requirement for this project
Eastern Cape Environmental Management Bill (Notice 205 of 2019)	The goals of the Eastern Cape Environmental Management Bill are as follow: • To supplement national legislation in the Province where necessary to protect the environment by providing reasonable measures for – ○ The management, protection and conservation of certain areas of ecological or environmental importance; ○ The promotion of the sustainable utilisation of the areas of ecological or environmental importance; ○ The management, protection and conservation of biological diversity and of the components of such biodiversity; and ○ The use of indigenous biological resources in a sustainable manner; • To provide for sound environmental management, enhancing and encouraging sustainable use of resources; • To encourage conservation, a risk averse approach; • To set provincial requirements, norms and standards for provincial environmental management according to best practices; and • To provide for compliance with the measures set out in paragraph (a); and • Generally, to give effect to section 24 of the Constitution within the Province.	The applicant must ensure that the project design, construction, and operation apply a precautionary, risk-averse approach, prioritise biodiversity protection, and demonstrate sustainable use of natural resources. The project must comply with any provincial environmental norms and standards and secure necessary provincial inputs/approvals in addition to national authorisations.
Eastern Cape Biodiversity Conservation Plan (BCP) (2019)	The primary intention of NEMBA Chapter 3 is to facilitate conservation and management of biodiversity in “biodiversity priority areas” or priority areas for conservation, outside of the Protected Area network, at a landscape level. The BCP may be used as a precursor for further refined mapping in Bioregional or Biodiversity Sector Plans undertaken at more localised scales (e.g. District or Local Municipalities). Therefore, the purpose of the BCP is to provide a map of these important biodiversity areas and develop associated land use management guidelines to inform:	The applicant must screen the development against CBAs/ESAs and avoid development in no-go biodiversity areas. Where avoidance is not feasible, impacts must be minimized and offset according to BCP guidelines. Rehabilitation and restoration plans must be implemented to maintain ecological connectivity.
MUNICIPAL		
Nelson Mandela Bay Bioregional Plan (2014)	The purpose of the final Bioregional Plan is to provide a map of biodiversity priorities and accompanying guidelines to inform land-use planning, environmental assessment and authorisations, and natural resource management by a range of sectors whose policies and decisions impact on biodiversity. The Bioregional Plan is a spatial plan that shows terrestrial and aquatic features that are critical for conserving biodiversity and maintaining ecosystem functioning. These areas are referred to as Critical Biodiversity Areas (CBAs).	The NMBM was used to screen the proposed development against CBAs and ESAs to identify biodiversity constraints.

DRAFT EMPR: WALMER HOUSING DEVELOPMENT ON ERF 11305

Legislation/ guidelines	Summary	Legal requirement for this project
Nelson Mandela Bay Municipality Municipal Spatial Development Plan (2025)	The strategic framework for the NMBM SDF provides an overarching guide for developing a long-term vision and actions to manage spatial development, balancing economic, social, and environmental goals through a coordinated process. It serves as a strategic plan that establishes the basis for more detailed spatial plans and their implementation.	The municipal SDF was consulted to ensure that the proposed development aligns with the spatial development objectives, land use guidelines and development vision of the municipality.

6 PLANNING AND DESIGN PHASE

This section of the EMPr provides management principles and mitigation measures for the pre-construction phase of the project. The environmental actions, procedures and responsibilities as required during the pre-construction phase are specified below.

Table 9. Impacts and Mitigation Measures in Planning and Design Phase

	Aspect	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
1.	Legal and social compliance	• Ensure environmental authorisations namely EA is obtained for the development. • All relevant plant and animal removal/relocation permits must be obtained from the competent authorities to remove any protected plant and animal species. • Environmental Control Officer (ECO) to be appointed to monitor compliance with permits conditions. • All health and safety permits and certification to be in place. • Appointment of a Community Liaison Officer (CLO). • Appointment of a Social Facilitator (SF) to manage social-economic aspects of the project relating to community and local business interest, particularly small, medium and micro-enterprise (SMMEs). • A health and safety plan in terms of the Occupational Health and Safety Act (Act No 85 of 1993) must be drawn up by the HSE officer prior to construction to ensure workers safety. • Environmental files to be compiled prior to commencement of construction activities. • Methods statement to be compiled for all works with environmental aspects and impacts, for approval of the ECO. • Employment and sub-contracting of locals..	• NMBM • Project Engineer	Once-off
2.	Layout issues	The development layout must conform to the following; • The business site to be designed such that there is no vehicular access from both the northern and southern portions of the development but that no through flow of vehicular traffic is possible. Separate parking areas to be provided for around the business sites.	• NMBM	Once-off

	Aspect	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
		• A pedestrian walkway along the edge of the business site to be provided, with suitable concrete bollards or similar barrier to prevent vehicular access via the pedestrian walkway • Zoning of the forest patches must be Open Space III and managed as for conservation usage. The buffers around the patches as well as other open spaces to be zoned as Public Open Space (POS) I • Stormwater ponds to be restricted to the edges of any forest patches and in accordance with any conditions related to any authorisation obtained in terms of the NFA from DFFE		
3.	Site establishment	• Develop a site establishment method statement • The Contractor must prepare a Construction Site Plan prior to establishing on-site. This plan must indicate: • The boundaries of the site that encompass all construction-related activities. • Vehicle and pedestrian access points and routes (safe diversions of routes when necessary). • Laydown area/s, offices, stockpile areas, storage areas, etc. • Construction site to be selected by the contractor in conjunction with the developer, consulting engineer and the ECO.	• Contractor	Once-off As required
4.	Existing civil services, infrastructure and properties	• Ensure that all existing services and infrastructure are known, mapped and marked to prevent potential damage. • Development of a plan to manage existing services to avoid damages • Wayleaves to be obtained from the relevant custodians for affected servitudes as applicable. • Property owners such as the local residents should be engaged throughout the project lifespan and engaged as necessary to promote conflict free development implementation. • Where relevant, the directly affected property owners should be informed in writing and engaged to reach a consensus and those proceedings and relevant agreements to be documented.	• Contractor	Once-off
5.	Water consumption	• Water conservation measures must be adopted into the building design. • Rainwater tanks should be considered.	• NMBM	Once-off

	Aspect	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
6.	Bulk services (Electricity and sewerage)	• A Bulk Services Report should be conducted. • The design of all residential dwellings, as well as community facilities, school buildings and business buildings should consider the use of alternative renewable energy sources such as CFL or LED lighting, solar geysers, solar street lighting etc. • All buildings should have a rainwater harvesting tanks. • Ensure that the municipality will have sufficient capacity for connection to existing infrastructure.	• NMBM	Once-off
7.	Stormwater management	• Infrastructure should be planned and designed in such a way as to take increased stormwater runoff in consideration.	• NMBM	Once-off

7 CONSTRUCTION PHASE

This section of the EMPr provides management principles and mitigation measures for the construction phase of the project. Environmental actions, procedures and responsibilities as required during the construction phase are specified. These specifications will form part of the contract documentation and therefore the Contractor will be required to comply with these specifications to the satisfactory of the Project Manager and Environmental Control Officer.

Table 10. Impact and Mitigation Measures in Construction Phase

	Aspect	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
1.	Legal and social compliance	- ECO to be appointed to audit environmental compliance with conditions in the EMPr and other permits and authorisations. - CLO and Social Facilitator to be retained throughout the construction tenure to represent the surrounding communities and local business, particularly SMMEs.	NMBM	Regularly
2.	Site establishment and security	- Fulltime security officers to be appointed to safeguard the site, its personnel, plant and equipment - Necessary health and safety as well as road construction signage to be installed. - Construction areas are to be clearly demarcated, and all construction activities are to be restricted to within these demarcated areas.	Contractor	Regularly
3	Water consumption	- A sustainable and lawful water supply should be utilised. - Consistent water supply for construction staff and activities. - If water is stored on-site, drinking water and multi-purposed water storage facilities shall be clearly distinguished and demarcated. - Implement rainwater harvesting techniques, where possible. - Re-use wash water for selected activities such as concrete batching.	Contractor	Regularly
4	Sanitation	- Ensure availability of ablutions for males and females. One ablution facility for every 10 workers during construction phase. - Ensure regular servicing of the ablution toilets by a service provider. - Ablution facilities to be located outside the regulated water area by all means practically possible.	Contractor	- Regularly - As required
5	Loss of indigenous vegetation	- Minimise vegetation clearance and confine to the proposed infrastructural footprint. - No blanket clearing of vegetation is to occur on the development site. Vegetation removal must be done in phases, to reduce the area of any any exposed surfaces and potential for dust pollution to occur at any one time.	Contractor	- Regularly - As required

	Aspect	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
		- When excavating, ensure that topsoil is separated from the subsoil to preserve the indigenous vegetation seedbanks of the area. - Use existing access roads as opposed to initiating new route, the latter should be authorised first prior to initiating. - Construction should not take place outside the authorised site footprint - Minimise vegetation clearance as practically possible to do so.		
6.	Disturbance of fauna and loss of habitats.	- A suitably qualified individual is to carry out a faunal "search and rescue", which fauna are to be relocated to a suitably protected area prior commencement of vegetation clearing. Should any injured fauna be found, they are to be taken to be examined by a veterinarian and if deemed suitable for rehabilitation, such are to be taken to a suitable faunal rehabilitation centre, in consultation with the department's biodiversity unit. Any species protected in terms of the Cape Nature and Environmental Ordinance (Ordinance 19 of 1974) which need to be removed or relocated require the necessary permits to be obtained from DEDEAT. - Any fauna or avi-fauna on site is to remain undisturbed as far as possible and shall not be trapped/killed or otherwise caught by any contractor or individuals carrying out any clearing or construction work. - No termite mounds should be intentionally destroyed. Any lizards, geckos, agamids, monitors or snakes encountered should be allowed to escape to a suitable habitat away from the disturbance. - The construction must be completed as quickly as possible - no fauna species may be killed. - Any bird nests that are found during the construction period must be reported to the Environmental Control Officer (ECO).	- NMBM - Contractor	- Once-off - Regularly
7.	Loss of SCC	- A plant search and rescue to be conducted by a suitably qualified botanical specialist prior to commencement of any vegetation clearing for each progressive stage of clearing for construction, for any protected species and species of special concern that may be affected by the development as well as other indigenous plant specimens which can be relocated. Such specimens are to be relocated and appropriately transplanted, where applicable and practical to the open space and forest patch conservation area. - Any protected species which need to be removed or relocated require the necessary permits to be obtained from DEDEAT for those species protected under the Cape Nature and Environmental Ordinance (Ordinance 19 of 1974). The relevant permits are to be obtained	- NMBM - Contractor - ECO	- Once-off - Regularly

	Aspect	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
		from the department of Forestry, Fisheries and the Environment for any species protected under the National Forest Act (Act 84 of 1998).		
8	Loss of animal SCC	- No animal SCC were identified. - Animals protected by provincial legislation may occur. These include local reptiles (snakes, lizards, geckos, tortoises and chameleons). All reptiles in the Eastern Cape are protected. - Permits (if required) must be obtained to relocate any protected animals identified. - Animals must only be relocated into the surrounding habitats. - Animals must be relocated by a trained and qualified handler as some of them may be venomous. - No poaching by construction staff will be allowed. - No killing of wild animals will be allowed.	Contractor	- Once-off - Regular
9.	Loss of biodiversity value	- All areas outside the demarcated construction footprint will be considered as No-Go areas. - Forest patches must be avoided. Permits must be obtained from Forestry to impact in any of the delineated forest patches.	Contractor	Regularly
10.	Habitat fragmentation	No areas of forest/thicket vegetation occurring outside the areas designated 'Open Space Zone 3 (conservation usage)' in the site layout plan/rezoning and subdivision plan should be lost because of the clearing.	Contractor	Regularly
11.	Impact on forest	- Authorisations/approvals as may be required in terms of the NFA should be obtained from DFFE - Forest patches should be no-go areas during construction.	Contractor	Regularly
12.	Stormwater and erosion impacts	- Minimise disturbance and clearing of vegetation - The site must be continuously monitored for erosion. - Appropriate erosion reducing methods must be implemented if required.	Contractor	Regularly
13.	Alien invasion proliferation	- Develop and implement an Alien Vegetation Management Plan to mitigate the establishment and spread of undesirable alien plant species during construction and operations. - Site must be continuously monitored and managed for alien vegetation growth. - Any alien invasive species which colonises disturbed ground must be systematically removed and destroyed prior to it attaining the seed formation stage. - Removal must occur through appropriate methods such as hand pulling, application of chemicals, cutting, etc. in accordance with the NEMBA: Alien Invasive Species Regulations.	Contractor	Regularly
14.	Waste management	- Develop a comprehensive solid waste management plan. - Provide clearly labelled waste bins with lids	Contractor	Regularly

	Aspect	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
		- Littering is strictly prohibited. - All waste generated on site shall be collected and appropriately disposed of at a registered municipal landfill site. - No on-site burning, burying or dumping of any waste materials, litter or refuse shall occur. - The Contractor shall identify and separate materials that can be re-used or recycled to minimise waste e.g. metals, packaging and plastics, and provide separate marked bins for these items. - Skips are to be provided for the temporary storage of construction waste. - All staff shall be trained on correct waste management. - Records of disposal of all waste generated on site shall be maintained for auditing purposes. - No wastewater shall be disposed of to the surrounding soil or natural water resources. - All effluent water from the camp/ office sites shall be disposed of in a properly designed and constructed system, situated so as not to adversely affect water sources. - All cement wastewater shall be collected in a container, allowed to evaporate, and the sludge disposed of as waste. Under no circumstances shall it be allowed to enter soil, groundwater resources or stormwater.		
15.	Hazardous material storage and handling	- No cement/concrete mixing is permissible on the soil surface. Cement mixers are to be placed on large trays to prevent accidental spills from coming into contact with soil surface. - Generators and fuel supply needed during construction must be placed on trays which rest on clean sand. Once construction has been completed, this sand must be removed from site and disposed of at a registered waste disposal site. - No servicing of vehicles and other machinery to take place on site and no fuel or other hazardous material to be stored on site.	Contractor	As required Regularly
16.	Traffic impact	- Ensure construction vehicles are visible and make use of Victoria Drive to gain access to site. - Pedestrians and motorists should be accommodated through appropriate traffic warning signs and measures. - Speed should be kept at a minimum next to the construction site.	Contractor	Regularly
17.	Visual impact	- The site camp must be clearly demarcated and screened in order to block any unappealing aesthetics within the site camp such material storage areas. - The construction footprint must be kept as small as possible, to avoid unnecessary disruption to the surrounding environment.	Contractor	Regularly

	Aspect	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
		- Disturbed areas to be rehabilitated / revegetated as soon as possible during the construction phase. - Areas utilised for construction at the site must be kept clean and tidy at all times.		
18.	Noise pollution	- Ensure that construction vehicles and working machinery are serviced and are in good condition to reduce their noise levels. - Construction activities should be maintained during the normal working hours (08h00-17h00). - Where works are to be carried outside the normal working hours, the adjacent residents affected must be informed, and such be undertaken within the shortest time possible. - Vehicles and plants to be inspected daily for defects which may cause elevated noise generation. - Vehicles and plants to be adequately serviced and maintained in good condition.	Contractor	Regularly as required
19.	Air pollution	- Implement dust suppression measures as required. - Vehicle speed should be kept at a minimal to reduce dust pollution. - Construction plant, equipment, machinery and vehicles should be well maintained and services regularly to minimise exhaust fumes air pollution. - Exposed areas should be re-vegetated as soon as all installations are completed.	Contractor	- Regularly - As required
20.	Paleontological and heritage resource disturbance	Archaeological: - A qualified archaeologist must conduct archaeological monitoring during all vegetation clearing activities and subsequent excavations associated with the proposed development. Monitoring is required due to the confirmed presence of archaeological human remains and associated cultural material along the north western boundary of the project area, as well as the broader heritage sensitivity of the Walmer dune landscape. Vegetation clearing and initial earthworks have the potential to expose additional burials, shell middens, stone artefact scatters, or other subsurface archaeological deposits that were not visible during the desktop update. - Monitoring must occur during: - Mechanical or manual vegetation removal, particularly in areas of dune deflation, natural hollows, or previously undisturbed terrain. - Topsoil stripping and bulk earthworks, including trenching, levelling, and foundation excavations.	Contractor	Regularly

	Aspect	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
		○ Any ground-disturbing activities within proximity to the previously exposed burial locality. ● If archaeological material, human remains, or burial related features are identified at any stage, all work in the immediate area must cease, and the finds must be reported to ECPHRA in accordance with Section 36 of the National Heritage Resources Act (Act 25 of 1999). Work may only resume once the heritage authority has issued written instructions or a permit authorising further action. ● This monitoring requirement must be incorporated into the contractor's method statements and the project's Environmental Management Programme (EMPr) to ensure compliance with NEMA and the NHRA. ● It is recommended that living heritage considerations be formally included as part of the public participation process in accordance with Sections 3(2)(b) and 38(3) of the National Heritage Resources Act (NHRA, Act 25 of 1999). Public consultation must provide affected communities, traditional leaders, and knowledge holders with the opportunity to identify culturally meaningful places, ritual spaces, ancestral associations, and other forms of living heritage that was not visible during field assessment. The archaeologist must be informed of any living heritage spaces that may be identified during the public participation process. These spaces must be identified and documented together with the affected communities who have identified them. The results of the consultation must be included as part of an updated or revised report including mitigation measures for these sites and possible no-go zones. ● Contractor teams must be briefed on heritage sensitivities, chance find procedures, and reporting requirements. Heritage awareness should form part of environmental induction and ongoing site compliance monitoring. ● All heritage related actions, including public participation, consultation, monitoring, chance find responses, and assessments—must be documented and included in the final heritage compliance report submitted to ECPHRA. ● If concentrations of pre-colonial archaeological heritage material, including subsurface dense artefact accumulations, associated cultural deposits, or human remains such as graves or burials—are uncovered during construction, all work must cease immediately and		

	Aspect	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
		the discovery must be reported without delay to the Eastern Cape Provincial Heritage Resources Agency (ECPHRA) at 043 492 1370. No further disturbance may occur until the heritage authority has provided guidance. Paleontological: • The Environmental Compliance Officer (ECO) responsible for overseeing the project should be well-informed about the potential presence of fossils within the project area. It is essential that they are aware of the likelihood of uncovering fossils during excavation into the underlying strata. • The construction managers and supervisory personnel involved in the project should also be made aware of the possibility of encountering significant fossils on-site. • During excavation, it is crucial that any geological exposures encountered are thoroughly inspected for the presence of fossil remains, particularly fossils of high scientific value e.g. vertebrate teeth and bones. If any fossils of significance are discovered, the ECO should contact the appropriate authority to further investigate the area. • In the event of significant fossil discoveries, a qualified palaeontologist should be promptly notified. This palaeontologist should possess the expertise and qualifications necessary for accurately assessing and documenting the findings. • Any fossils of scientific value found within the excavation area should be regarded as of palaeontological interest. The palaeontologist should be responsible for recording and sampling these fossils. The costs associated with this activity should be borne by the developer as part of their responsibility for preserving and managing palaeontological resources. • If a new paleontological resource is discovered, it is the jurisdiction of the SAHRA to recover and store such resource. Hence, further exploration must be conducted by a professional palaeontologist		
21.	Employment opportunities	• Employment of local labour and contracting of SMMEs.	• Contractor	• Regularly

8 REHABILITATION PHASE

Post construction obligations, also application during works, of the project are outlined in the subsequent table to ensure that contractor leaves site in an acceptable condition with minimal environmental impacts.

Table 11. Impact and Mitigation Measures in Operational Phase

	Activity/Impact	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
1.	Site de-establishment	All structures should be removed and the site must be clean with no waste, rubble or construction material.	Contractor	Once-off
2.	Revegetation	Rehabilitation of vegetation on the site must be done as described in the Rehabilitation Plan.	Contractor	Regularly

9 OPERATIONAL PHASE

This section of the EMPr provides management principles for the operational phase of the project. Environmental actions, procedures and responsibilities as required during the operational phase are specified.

Table 12. Impact and Mitigation Measures in Operational Phase

	ASPECT	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
1.	Open spaces and Forest patches management	- No forest patches are to be developed or impacted on. These sites are to be appropriately zoned to Open Space III for conservation purposes and the protection thereof and must be incorporated into the NMBM's Metropolitan Open Space Management Plan/Bioregional Plan and managed as a conservation area. - The forest patches must be fenced off with a suitable fencing solution such as "Bettafence" to prevent dumping and destruction of the forest. - Regular inspections of the fenced areas are to be made to monitor whether dumping of rubble or refuse is occurring.	NMBM	Ongoing
2.	Waste management	Regular (weekly) waste collection service to be provided	NMBM	Ongoing

	ASPECT	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
3.	Freshwater utilization	- Water-wise practices should be considered such as use of rainwater for non-portable water requirements. - Leak monitoring, detection repairs be implemented.	NMBM	As required
4.	Bulk services	- Consider use of alternative energy sources such as solar panels placed on the infrastructure roofs to power the facility. - The municipality must ensure sewer system is not overloaded. - Regular maintenance of the system required.	NMBM	Ongoing As required
5.	Stormwater management	- Once the storm water facilities have been completed, and fenced to prevent access by the public, the maintenance and monitoring will remain the responsibility of the municipality. These are some of the suggested Best Management Practices (BMPs) for the development - Public open space – Vegetation to be maintained such that flood water levels are not increased. - Storm water structures – Ponds frequently monitored and obstructions (debris) to inlet/outlet works, removed. - Open Channels – Channels to be regularly monitored to identify any erosion gullies or silt deposition. - Road Construction – No concentrated flow points are allowed onto road area. - Pipes – All pipes are to be frequently monitored and cleared of obstructions. - Develop a comprehensive stormwater management program which addresses management of stormwater detention ponds, the discharge outlets, inclusive of erosion protection measures and mitigation of such erosion occur. - Include periodical site inspection that inspects the effectiveness of the run-off control system and specifically records occurrence or not of any erosion on site or downstream. - Outlet structures at road culverts to be designed in a manner that dissipates flow energy. - Vegetation to be maintained in public open space such that flood water levels are not increased. - Ponds frequently monitored and obstructions (debris) to inlet and outlet works, removed.	NMBM	Ongoing As required

	ASPECT	RECOMMENDED MITIGATION MEASURES/ MANAGEMENT ACTIONS	RESPONSIBLE PARTY	FREQUENCY OF ACTION
		Channels to be regularly monitored to identify any erosion gullies or silt deposition.		
6.	Increase in traffic and pedestrian levels in the area	Upgrade intersections; provide additional facilities for public transport and vulnerable road users (pedestrians and cyclists).	NMBM	As required
7.	Neighbouring upmarket houses value impacts	- No deviating from definite pricing contours in the layout. - The area directly adjacent Walmer Heights must be reserved for higher income households, with lower priced housing along Victoria Drive	NMBM	Ongoing
8.	Increased water demand	- Implement water-efficient building design. - Promote water conservation awareness among residents.	NMBM	Ongoing

10 ROLES AND RESPONSIBILITIES

10.1 PROJECT MANAGER

The Project Manager is responsible for overall management of the project and the implementation of the EMPr. The following tasks fall within his / her responsibilities:

- Be familiar with the recommendations and mitigation measures of this EMPr, and implement these measures;
- Monitor site activities on a daily basis for compliance;
- Conduct internal audits of the construction site against the EMPr;
- Confine the construction site to the demarcated areas; and
- Rectify transgressions through the implementation of corrective action

10.2 ENVIRONMENTAL CONTROL OFFICER

The Developer must appoint a suitable qualified Environmental Control Officer that will be responsible for ensuring compliance to the EMPr. The Environmental Control Officer on site will conduct regular site visits to ensure the success of the EMPr.

The Environmental Control Officer will:

- Do regular site visit to be able to compile site inspection reports for the inclusion in the EMPr as an addendum if necessary.
- Report compliance and non-compliance to the competent authority.
- Know the contents and implications of the environmental report and monitor the implementation of the findings using the EMPr.
- Act as a guide, advisor and consultant to the contractor and client on environment issues during construction. This will be achieved by continuous auditing of the project, identification of problem areas and provisioning of action plans to avoid costly stoppages and /or environmental damage.
- Ensure that people can report incidents and recommend corrective action when required for aspects of non-compliance within the EMPr
- Upgrade the EMPr as necessary and inform the relevant parties of the changes.

10.3 CONTRACTOR

The construction contractor is responsible for submitting reports on implementation of the environmental plan as described by the project proponent and authorities. The contractor must always ensure compliance with the EMPr

during construction activities and maintain an environmental register which keeps a record of all environmental incidents that occur such as

- Public involvement / complaints;
- Health and safety incidents;
- Incidents involving Hazardous materials stored on site; and
- Non-compliance incidents.

11 IMPLEMENTATION OF ENVIRONMENTAL MANAGEMENT PROGRAMME

A copy of the EMPr must be kept on site at all times during the construction period. The EMPr will be binding on all contractors operating on the site and must be included within the Contractual Clauses. The monitoring programme must be implemented for the duration of the construction and operation of the erosion control structures. This programme should include:

- Establishing a baseline of pre-construction site conditions validated with photographic evidence.
- Monthly audits will be conducted by an independent ECO for the construction phase to ensure compliance with the conditions stipulated in this EMPr and, where necessary, make recommendations for corrective action. These audits can be conducted randomly and do not require prior arrangement with the Project Coordinator.
- Compilation of an audit report with a rating of compliance with the EMPr. The ECO must keep a photographic record of the demarcated site and construction area. The Contractor must be held liable for all unnecessary damage to the environment. A register must be kept of all complaints from the community. All complaints must be handled immediately to ensure timeous rectification by the responsible party.

11.1 ENVIRONMENTAL AUDITS AND MONITORING

The construction contractor will submit reports on implementation of the environmental plan as described by the project proponent and authorities. An environmental audit will be conducted prior to the development and construction activities. This environmental audit will ensure that:

- Mitigation measures are implemented as prescribed in the Environmental Management Programme;
- The relevant authorities are kept informed about progress with the project and that they are given assurance that the project is implemented and managed as prescribed in the EMPr.
- Periodic inspections and/or audits are performed.
- Compliance to the conditions of the EMPr is adhered to and a report compiled.
- Reviews of conformance against policies and procedures stated in this document are performed.
- Supervisors in all work areas will conduct performance and compliance reviews, using the EMPr as guideline to ensure compliance.

- Inspections will occur monthly (or as required).

11.2 RECORD KEEPING

Documents to be maintained by the designated representative/ site agent are to include:

- Training records.
- Inspection records.
- Records of non-compliance and corrective action.
- Records of all complaints, concerns or issues and corrective action.
- Environmental Management Programme.
- All incidents reports.

All records will be kept for up to a year after the completion of the project or in accordance with other legal requirements as they apply.

11.3 EMPR UPDATES

The EMPr will be subject to on-going review throughout the course of the project to ensure its continued suitability, adequacy, and effectiveness. This review may include but will not be limited to monitoring and measuring information, performance data, assessment and audit results and other relevant information and data.

11.4 NON-COMPLIANCE

The contractors must act immediately when notice of non-compliance is received and take corrective action. Complaints received regarding activities on the construction site pertaining to the environment must be recorded in a dedicated register and the response(s) noted with the date and action taken. The ECO should be made aware of any complaints.

Any non-compliance with the agreed procedures of the EMPr is a transgression of the various statutes and laws that define how the environment is managed. Failure to redress the cause must be reported to the competent authority for them to deal with the transgression, as it deems fit. The Contractor is deemed not to have complied with the EMPr if, inter alia:

- There is evidence of contravention of the EMPr specifications within the boundaries of the construction site and site extensions;
- There is contravention of the EMPr specifications which relate to activities outside the boundaries of the construction site;
- Environmental damage ensues due to negligence;
- Construction activities take place outside the defined boundaries of the site; and/or
- The Contractor fails to comply with corrective or other instructions issued within a specific time.

It is recommended that the Contractors institute penalties for the following less serious violations and any others determined during work, as detailed below:

- Littering on site.
- Lighting of illegal fires on site.
- Persistent or unrepaired fuel and oil leaks.
- Any persons, vehicles or equipment related to the Contractor's operations found within the designated "no-go" areas.
- Excess dust or excess noise emanating from site.
- Possession or use of intoxicating substances on site.
- Any vehicles being driven more than designated speed limits.
- Removal and/or damage to fauna, flora, cultural or heritage objects on site.
- Urination and defecation anywhere except at designated facilities.

11.5 INCIDENT REPORTING AND REMEDY

If a leakage or spillage of hazardous substances occurs on site, the local emergency services must be immediately notified of the incident. The following information must be provided:

- the location;
- the nature of the load;
- the extent of the impact; and
- the status at the site of the accident itself (i.e., whether further leakage is still taking place, whether the vehicle or the load is on fire).

Written records must be kept on the corrective and remedial measures decided upon and the progress achieved therewith over time. Such progress reporting is important for monitoring and auditing purposes. The written reports may be used for training purposes to prevent similar future occurrences.

11.6 PENALTIES

Where environmental damage is caused or a pollution incident, and/or failure to comply with any of the environmental specifications contained in the EMPr, the Developer and/or the Contractor will be liable. The following violations, and any others determined during work, should be penalised:

- Hazardous chemical/oil spill and/or dumping in non-approved sites.
- Damage to sensitive environments.
- Damage to cultural and historical sites
- Unauthorised removal/damage to indigenous trees and other vegetation, particularly in identified sensitive areas.

- Uncontrolled/unmanaged erosion.
- Unauthorised blasting activities (if applicable).
- Pollution of water sources.
- Unnecessary removal or damage to trees.

12 SUMMARY OF RECOMMENDATIONS AND CONCLUSION

The proposed development will have minimal environmental impacts which are manageable through good engineering practices and following all environmental recommendations prescribed. Although all foreseeable actions and potential mitigations or management actions are contained in this document, the EMPr should be considered as a day-to-day management document. The EMPr thus sets out the environmental standards that are required to minimize the negative impacts and maximize the positive benefits of the local communities. An EMPr is a “live document” and its continuous review and correct management will definitely result to the successful construction of the proposed development.

The contractor is to be made aware of the potential cost and timing implications needed to fulfill the implementation of the EMPr, thus adequately costing for these.

Annexure 1: Rehabilitation Schedule and Guideline

Where applicable, the rehabilitation schedule and procedure to be adhered to is as follows (this methodology should be updated during the construction phase):

Table 13. Rehabilitation schedule and guideline

Step	Method	Equipment
1	Remove all construction material from the area where construction has been completed	To be undertaken by hand.
2	The ground should be sloped so as to attain a natural slope and to attain a natural water flow, if it has been altered during construction (the natural slope should be altered as little as possible during construction).	To be undertaken by hand.
3	Topsoil that has been stockpiled during construction must be applied to the area to undergo rehabilitation. The depth of the topsoil layer to be applied depends on the natural depth of topsoil in the area, and the amount of topsoil that may have been lost during construction.	Topsoil must be applied from the topsoil stockpiled during construction
4	The area should be mulched to improve water retention, and brushwood applied to act as a soil stabiliser. Mulch and brushwood must be applied more heavily in areas which are presently well-wooded (or as directed by the ECO).	The mulch used should be woodchip, obtained commercially or from trees removed during site clearance. The brushwood is obtained from the bushes and trees removed during site clearance.
5	The planting of locally indigenous species should be undertaken, with local species endemic to the area.	The plant species to be used during the rehabilitation process should be endemic to the area surrounding the substation installation.
6	The species which has been planted should be regularly watered after planting. Watering should ensure that no erosion of the topsoil takes place.	A hosepipe must be available on site.
7	Plant species which are not well established after a period of 2 months after planting should be re-planted. If necessary, another dressing of topsoil should be applied prior to planting.	As above
8	Slope stabilisation measures may be necessary in areas where vegetation has not been able to establish and there is an erosion risk. The measures implemented depend on the situation, and can be varied as necessary. The ECO will direct the Contractor as required.	Various slope stabilisation measures are available and vary in effectiveness according to the situation including • Onion bags • Soil saver (jute/hessian) blankets • Logs/bark held in place with pegs

Annexure 2: Proposed Fines and Penalty Structure

Table 14. Proposed fines and penalty structure

Typical incidents incurring penalties	Value
Failure to secure construction site from public access	R5,000
Failure to demarcate working areas and servitudes and/or maintain fences and/or demarcation tape.	R1,000
Failure to stockpile topsoil correctly (per incident)	R2,000
Failure to stockpile materials in designated areas (per incident)	R500
Discharging effluent and/or polluted storm water onto the ground or into surface water (per incident)	R2,000
Failure to provide adequate sanitation, waste disposal facilities or services (per incident)	R1,000
Failure to demarcate construction area boundaries before commencing construction clearance and other activities (per incident)	R5,000
Venturing into or undertaking construction related activities within no-go areas, without formal written approval from the ECO (per incident)	R5,000
No induction regarding environmental matters and site housekeeping practices (per employee)	R2,000
Stockpile of soils and materials outside demarcated areas (per incident)	R1,000
Inappropriate mixing of cement/concrete and poor management of concrete slurry (per incident)	R2,000
Burning of waste on site (including cement bags) (per incident)	R 2,000
Untidiness and litter at camp (per incident)	R200
Unauthorised removal of indigenous trees, medicinal or other plants (per incident)	R2,000
Damaging/killing animals/birds (per incident)	R 1,500
Failure to erect temporary fences as required (per incident)	R2,000
Failure to reinstate disturbed areas within the specified timeframe (per incident)	R2,000
Fire – costs of runaway fires will be borne by the Contractor, should he/she be proven responsible for such fires (per incident)	R25,000
Failure to provide adequate equipment for emergency situations (per incident)	R5,000
Defacing, painting or damaging natural or heritage features (per incident) – mandatory removal of employee from site	R5,000
Damaging cultural, historical and/or archaeological sites of importance (per incident) – mandatory removal of employee from site	R5,000
Failure to maintain basic safety measures on site	R1,000
Failure to carry out required community liaison, damage to property etc., without prior negotiation and/or compensation and other social infringements (per incident)	R1,000
Persistent and un-repaired oil leaks from machinery. The use of inappropriate methods of refuelling (per incident)	R2,000
Failure to provide drip trays and/or empty them frequently (per incident)	R500
Inappropriate use of bins and poor waste management on site (per incident)	R500
Inappropriate off-site disposal of waste from site (per incident)	R10,000
Deliberate lighting of illegal fires on site (per incident)	R1,000
The eating of meals on site outside the defined eating area. Individual not making use of the site ablution facilities (per incident)	R200
Inappropriate use of adjacent watercourses and water bodies – such as for unapproved water abstraction, washing of vehicles, wastewater disposal and use by employees for washing (per incident)	R1000
Any person, vehicle, item of plant, or anything related to the Contractor's operations causing a public nuisance (per incident)	R500
Construction vehicles not adhering to speed limits (per incident)	R200
Failure to maintain and register incidents in the incident register (per incident)	R1,000
Failure to remove all temporary features and leftovers from the construction site and works areas upon completion of the works (per incident)	R50,000
Any contravention with a Method Statement (per incident)	R5,000
Repeated contravention of the specifications or failure to comply with instructions (per incident)	R5,000

Annexure 3: Issues to be addressed by and where appropriate shown on the Environmental Management Site Plan

Table 15. Issues addressed on the EMP Site Plan

Issue	Nature / Description
Sequence of events	Description of the nature of the process required. Briefly describe the sequence of events that will take place from the time that the contractor moves onto site to the time when the site is handed over to the Project Proponent
Health and safety	Potential risks and hazards and precautions that will be taken Cooking area, hazardous materials site, first aid kit, fuel store, security issues, fire management, and public safety.
On site toilets	How many required for the particular development? Are there existing ablution facilities that can be used? If not, how many toilets need to be brought to site and how long are the toilets required on site? Location of toilets (Site Plan)
Workforce	Number of onsite workers Training of workforce in terms of environmental awareness Management of workforce, particularly sub-contractors
Transport and traffic	Transport required for site workers Routes to be used by construction vehicles Demarcate location of traffic turning circle and parking areas (Site Plan)
Infrastructure and associated equipment	Nature and extent of infrastructure needed for construction Indicate on site plan
Material storage (e.g. sand needed to build wall)	Approximation of quantity to be excavated Where to be stockpiled (Site Plan) How long to be stockpiled Area required for stockpile
Earthworks/cleaning	Volume of material to be excavated/cleaned Duration of operations Where stocks to be kept on site (Site Plan) How long to be kept on site Where, when and how to be disposed of Recycling and/or re-use of materials
Equipment needed for construction activities	Area required for material and equipment storage Duration of works Nature of equipment and necessary materials
Drinking water	Quantity required Duration of period in which required Source of water Location of potable water (Site Plan)
Cooking and rest areas	Area required Equipment required e.g. gas stoves, matches etc. Shelter Location – indicate on site plan
Existing structures	Indication of location of any structures that need to be removed and/or protected
Life of project	Working hours Time frame Phasing of work areas

Issue	Nature / Description
Construction site	Work area required Location of construction site and work area (Site Plan) Demarcation of no-go areas
Environmentally sensitive areas and possible environmental risks associated with construction activities	A plan of environmentally sensitive areas must be drawn up and made available to contractors. This will include the riverine corridor and pan demarcated in the field survey. A training programme on possible environmental risks that may result from construction activities and how to deal with these (including a reporting structure) must be made available prior to construction commencing.
Waste management	Litter drums - number, type, size, location (Site Plan) Closest registered waste disposal site (Location map) Waste management plan Recycling / material re-use options

Annexure 4: General Code of Conduct for Contractors

Contractors shall ensure that all sub-contractors, employees, suppliers, agents, etc., are fully aware of the environmental issues detailed in the Construction Environmental Management Programme.

Contractors must Investigate and comply with all existing regulations and laws/ bylaws unless the Relevant Authority grants specific written authority waiving compliance with any legislation.

Contractor's, *inter alia*, commit to the following code of conduct:

- Implementing, documenting, reviewing and complying with environmental specifications, permits and management plans as well as relevant environmental legislation, regulations and standards.
 - Ensuring the communication and training of all personnel through induction and appropriate environmental training.
 - Reducing and, where possible, eliminating and preventing any harmful effects to the environment and societal milieu
 - Liaising openly with, and reporting publicly to, all interested and affected parties on environmental matters.
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Annexure 5: Environmental Incident Register

Table 16. Environmental incident register

ENVIRONMENTAL INCIDENT REGISTER				
Date	Environmental incident	Comments <i>(Include any possible explanations for current condition and possible responsible parties. Include photographs, records etc. if available)</i>	Corrective Action Taken <i>(Give details and attach documentation as far as possible)</i>	Signature

Annexure 6: Environmental Complaints Register

Table 17. Environmental complaints register

ENVIRONMENTAL COMPLAINTS REGISTER				
Date	Environmental Complaint	Name of Compliance <i>(Name of complained and method which complaint was received)</i>	Corrective Action Taken <i>(Give details and attach documentation as far as possible)</i>	Signature

